

Exercise-Introduction To Version Control

Q1)Git Setup <https://confluence.atlassian.com/bitbucket/set-up-git-744723531.html>

```
harsh-raj@TTNPL-harsh:~$ sudo apt install git
[sudo] password for harsh-raj:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 31 not upgraded.
harsh-raj@TTNPL-harsh:~$ git --version
git version 2.43.0
```

Q2)Initialize a Git Repository

```
harsh-raj@TTNPL-harsh:~$ mkdir myproject
harsh-raj@TTNPL-harsh:~$ cd myproject
harsh-raj@TTNPL-harsh:~/myproject$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/harsh-raj/myproject/.git/
```

Q3)Add files to the repository

```
harsh-raj@TTNPL-harsh:~/myproject$ touch file.txt
harsh-raj@TTNPL-harsh:~/myproject$ echo "hello world" > file.txt
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file.txt

nothing added to commit but untracked files present (use "git add" to track)
harsh-raj@TTNPL-harsh:~/myproject$ git add file.txt
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file.txt
```

Q4)Unstage 1 file

```
harsh-raj@TTNPL-harsh:~/myproject$ git reset world.txt
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        world.txt
```

Q5)Commit the file

```
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   file.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    world.txt

harsh-raj@TTNPL-harsh:~/myproject$ git commit -m "Initial Commit"
[master (root-commit) d7de75c] Initial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
```

Q6)Add a remote

```
nothing added to commit but untracked files present (use "git add" to track)
harsh-raj@TTNPL-harsh:~/myproject$ git remote add origin https://github.com/HrshRaj30062004/Exercise
harsh-raj@TTNPL-harsh:~/myproject$ git remote -v
origin  https://github.com/HrshRaj30062004/Exercise (fetch)
origin  https://github.com/HrshRaj30062004/Exercise (push)
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    world.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Q7)Undo changes to a particular file

```
harsh-raj@TTNPL-harsh:~/myproject$ echo "changes made" >> world.txt
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   world.txt

no changes added to commit (use "git add" and/or "git commit -a")
harsh-raj@TTNPL-harsh:~/myproject$ git restore world.txt
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master
nothing to commit, working tree clean
```

Q8)Push changes to Github

```
harsh-raj@TTNPL-harsh:~/myproject$ git push -u origin master
Username for 'https://github.com': ^C
harsh-raj@TTNPL-harsh:~/myproject$ git push -u origin master
Username for 'https://github.com': HrshRaj30062004
Password for 'https://HrshRaj30062004@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 483 bytes | 483.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/HrshRaj30062004/Exercise
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Q9)Clone the repository

```
harsh-raj@TTNPL-harsh:~/myproject$ git clone https://github.com/HrshRaj30062004/Exercise.git
Cloning into 'Exercise'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
harsh-raj@TTNPL-harsh:~/myproject$
```

Q10) Add changes to one of the copies and pull the changes in the other.

```
harsh-raj@TTNPL-harsh:~/myproject$ echo "update from master repo" >> update.txt
harsh-raj@TTNPL-harsh:~/myproject$ git add update.txt
harsh-raj@TTNPL-harsh:~/myproject$ git commit -m "Update from origin master"
[master 2c66733] Update from origin master
 1 file changed, 1 insertion(+)
 create mode 100644 update.txt
harsh-raj@TTNPL-harsh:~/myproject$ git push
Username for 'https://github.com': HrshRaj30062004
Password for 'https://HrshRaj30062004@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 343 bytes | 343.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/HrshRaj30062004/Exercise
 92667aa..2c66733  master -> master
harsh-raj@TTNPL-harsh:~/myproject$ cd Exercise
harsh-raj@TTNPL-harsh:~/myproject/Exercise$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 323 bytes | 161.00 KiB/s, done.
From https://github.com/HrshRaj30062004/Exercise
 92667aa..2c66733  master      -> origin/master
Updating 92667aa..2c66733
Fast-forward
 update.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 update.txt
```

Q11) Check differences between a file and its staged version

```
harsh-raj@TTNPL-harsh:~/myproject$ echo "new line" >> file.txt
harsh-raj@TTNPL-harsh:~/myproject$ git add test.txt
fatal: pathspec 'test.txt' did not match any files
harsh-raj@TTNPL-harsh:~/myproject$ git add file.txt
harsh-raj@TTNPL-harsh:~/myproject$ echo "another line" >> file.txt
harsh-raj@TTNPL-harsh:~/myproject$ git diff --staged
diff --git a/file.txt b/file.txt
index 3b18e51..79ee69e 100644
--- a/file.txt
+++ b/file.txt
@@ -1 +1,2 @@
 hello world
+new line
```

Q12)Ignore a few files to be checked in

```
harsh-raj@TTNPL-harsh:~/myproject$ echo "ignore" >> ignore.txt
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   .gitignore
    new file:   Exercise
    modified:   file.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    ignore.txt

harsh-raj@TTNPL-harsh:~/myproject$ vi .gitignore
harsh-raj@TTNPL-harsh:~/myproject$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   .gitignore
    new file:   Exercise
    modified:   file.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   .gitignore
```

Q13)Create a new branch

```
harsh-raj@TTNPL-harsh:~/myproject$ git checkout -b vaibhav
Switched to a new branch 'vaibhav'
harsh-raj@TTNPL-harsh:~/myproject$ git branch
  master
* vaibhav
```

Q14)Diverge them with commits

```
harsh-raj@TTNPL-harsh:~/myproject$ git branch
  master
* vaibhav
harsh-raj@TTNPL-harsh:~/myproject$ echo "diverge branches" >> branch.txt
harsh-raj@TTNPL-harsh:~/myproject$ git add branch.txt
harsh-raj@TTNPL-harsh:~/myproject$ git commit -m " added diverged branches"
[vaibhav 2e3e1ee] added diverged branches
4 files changed, 7 insertions(+)
create mode 100644 .gitignore
create mode 160000 Exercise
create mode 100644 branch.txt
```

```
harsh-raj@TTNPL-harsh:~/myproject$ git checkout master
warning: unable to rmdir 'Exercise': Directory not empty
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
harsh-raj@TTNPL-harsh:~/myproject$ echo "diverge branches master" >> branch.txt
harsh-raj@TTNPL-harsh:~/myproject$ git add branch.txt
harsh-raj@TTNPL-harsh:~/myproject$ git commit -m " add change on master branch"
[master 8a98911] add change on master branch
1 file changed, 1 insertion(+)
create mode 100644 branch.txt
```

Q15)Edit the same file at the same line on both branches and commit

```
harsh-raj@TTNPL-harsh:~/myproject$ cat git.txt
cat: git.txt: No such file or directory
harsh-raj@TTNPL-harsh:~/myproject$ cat branch.txt
diverge branches master
harsh-raj@TTNPL-harsh:~/myproject$ git checkout vaibhav
Switched to branch 'vaibhav'
harsh-raj@TTNPL-harsh:~/myproject$ cat branch.txt
diverge branches
```


Q16)Try merging and resolve merge conflicts

```
harsh-raj@TTNPL-harsh:~/myproject$ git checkout master
warning: unable to rmdir 'Exercise': Directory not empty
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)
harsh-raj@TTNPL-harsh:~/myproject$ git merge vaibhav
Auto-merging branch.txt
CONFLICT (add/add): Merge conflict in branch.txt
Automatic merge failed; fix conflicts and then commit the result.
harsh-raj@TTNPL-harsh:~/myproject$ vi branch.txt
harsh-raj@TTNPL-harsh:~/myproject$ git commit -m "resolving conflict"
U      branch.txt
error: Committing is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.
harsh-raj@TTNPL-harsh:~/myproject$ git add branch.txt
harsh-raj@TTNPL-harsh:~/myproject$ git commit -m "resolving conflict"
[master 039e654] resolving conflict
harsh-raj@TTNPL-harsh:~/myproject$ git merge vaibhav
Already up to date.
```

Q17)Stash the changes and pop them

```
harsh-raj@TTNPL-harsh:~/myproject$ git stash
No local changes to save
harsh-raj@TTNPL-harsh:~/myproject$ echo "Temporary Text" >> branch.txt
harsh-raj@TTNPL-harsh:~/myproject$ git stash
Saved working directory and index state WIP on master: 039e654 resolving conflict
harsh-raj@TTNPL-harsh:~/myproject$ git stash list
stash@{0}: WIP on master: 039e654 resolving conflict
harsh-raj@TTNPL-harsh:~/myproject$ git stash pop
On branch master
Your branch is ahead of 'origin/master' by 4 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   branch.txt

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (007138e5b966827ae044e09fd114744e695689fd)
```


Q18) Add the following code to your .bashrc file :
color_prompt="yes" parse_git_branch()
{ git branch 2> /dev/null | sed -e '/^[^*]/d' -e 's/* \(.*/(\1)/' } if ["\$color_prompt" = yes];
then PS1='\u@\h\[\033[00m\]:\[\033[01;34m\]\W\[\033[01;31m\]
\$(parse_git_branch)\[\033[00m\]\\$ ' else PS1='\u@\h:\W \$(parse_git_branch)\\$ ' fi unset
color_prompt force_color_prompt

```
harsh-raj@TTNPL-harsh:~/myproject$ vi ~/.bashrc
harsh-raj@TTNPL-harsh:~/myproject$ source ~/.bashrc
bash: /home/harsh-raj/.bashrc: line 119: syntax error near unexpected token `('
bash: /home/harsh-raj/.bashrc: line 119: `color_prompt="yes" parse_git_branch()'
harsh-raj@TTNPL-harsh:~/myproject$ vi ~/.bashrc
harsh-raj@TTNPL-harsh:~/myproject$ source ~/.bashrc
harsh-raj@TTNPL-harsh:myproject (master)$
```